Data Mining Lab 1
Name:Sameer Gupta
Roll No:252010036

February 17, 2026

[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
```

[3]:
```
#Q1. Load Iris Dataset
iris = load_iris()

df = pd.DataFrame(
    data=iris.data,
    columns=iris.feature_names
)

df['species'] = iris.target
df['species'] = df['species'].map(dict(zip(range(3), iris.target_names)))

df
```

[3]:
```
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                  5.1               3.5                1.4               0.2
1                  4.9               3.0                1.4               0.2
2                  4.7               3.2                1.3               0.2
3                  4.6               3.1                1.5               0.2
4                  5.0               3.6                1.4               0.2
..                 …                 …                  …                 …
145                6.7               3.0                5.2               2.3
146                6.3               2.5                5.0               1.9
147                6.5               3.0                5.2               2.0
148                6.2               3.4                5.4               2.3
149                5.9               3.0                5.1               1.8

       species
0       setosa
1       setosa
2       setosa
3       setosa
```

```
4       setosa
..         …
145  virginica
146  virginica
147  virginica
148  virginica
149  virginica

[150 rows x 5 columns]
```

[4]:
```python
#Q2. Give the dimension of the data
df.shape
```

[4]:
```
(150, 5)
```

[5]:
```python
#Q3. Give the number of classes in the dataset and what are they(list them)?
df['species'].unique()
```

[5]:
```
<StringArray>
['setosa', 'versicolor', 'virginica']
Length: 3, dtype: str
```

[6]:
```python
#Q4. Each object in the data has how many features and what are they(list them)?
iris.feature_names
```

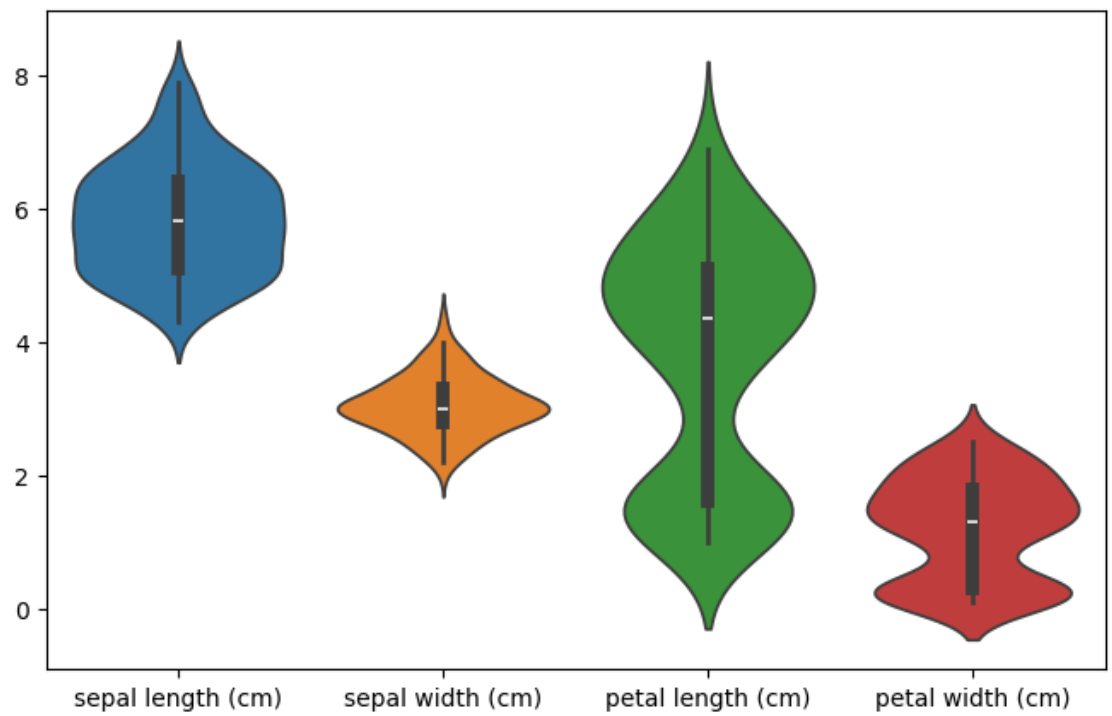
[6]:
```
['sepal length (cm)',
 'sepal width (cm)',
 'petal length (cm)',
 'petal width (cm)']
```

[7]:
```python
#Q5. Describe the data.(5 point summary)
df.describe()
```

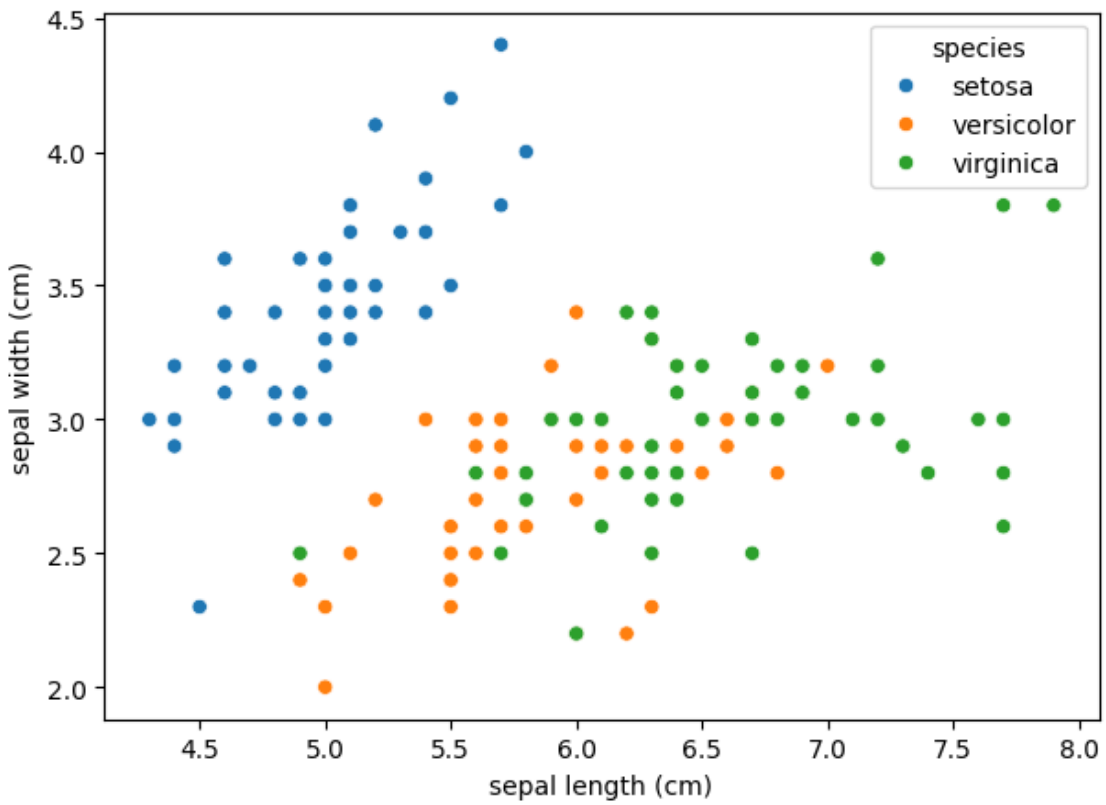
[7]:
```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)
count        150.000000
mean           1.199333
std            0.762238
min            0.100000
```

```
25%            0.300000
50%            1.300000
75%            1.800000
max            2.500000
```

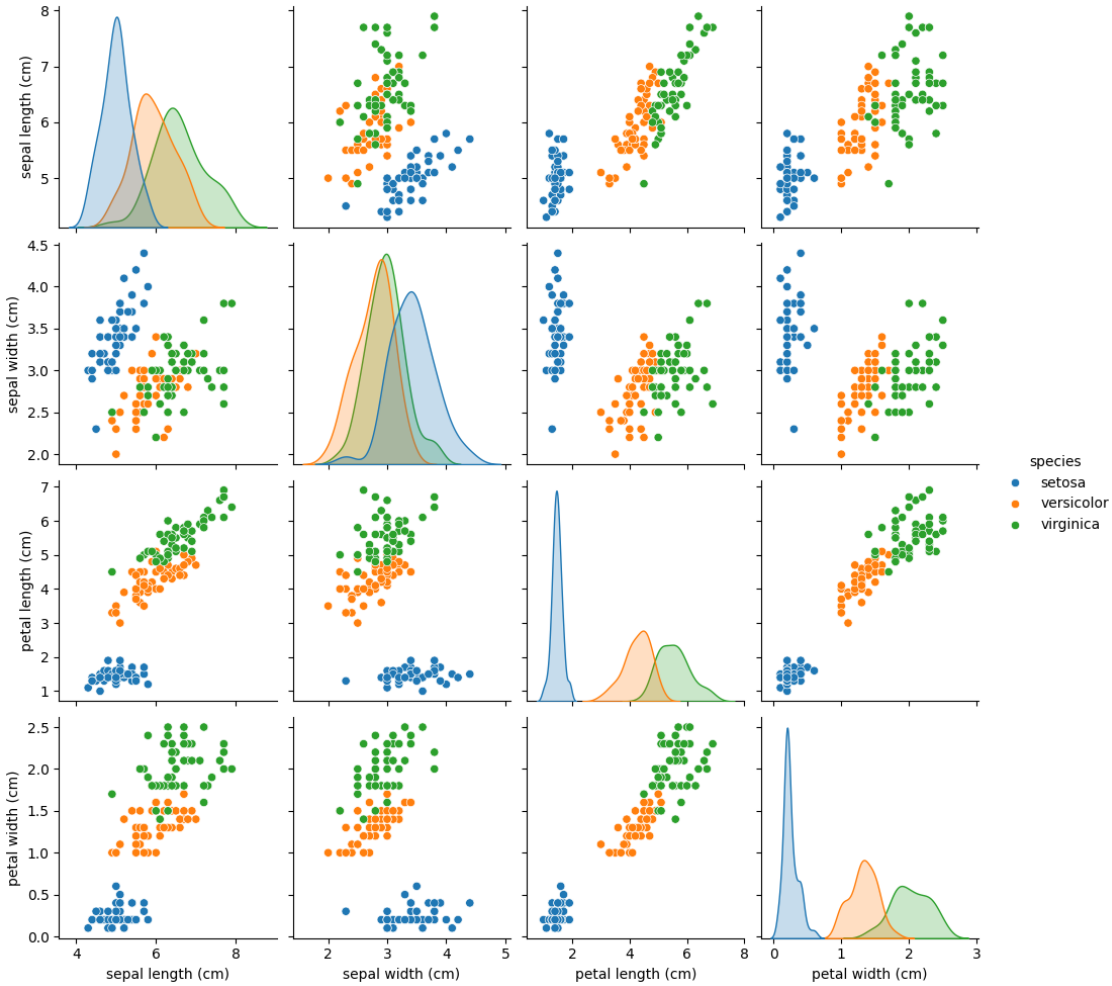
[8]:
```
#Q6. Visualize the data using 'violin plot'

plt.figure(figsize=(8,5))
sns.violinplot(data=df)
plt.show()
```



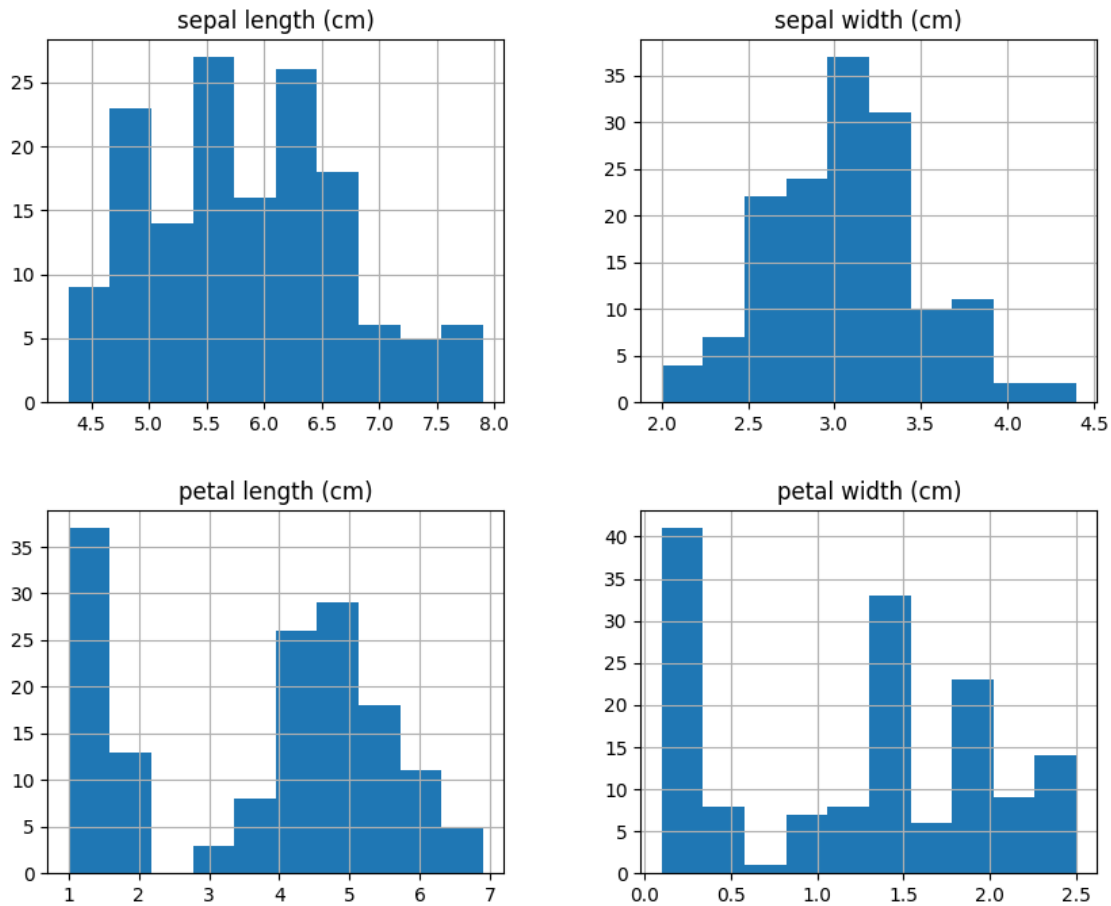
[9]:
```
#Q7. Visualize the data using 'scatter plot'
plt.figure(figsize=(7,5))
sns.scatterplot(
    data=df,
    x='sepal length (cm)',
    y='sepal width (cm)',
    hue='species'
)
plt.show()
```



[10]:
```
#Q8. Visualize the data using 'pair plot'
sns.pairplot(df, hue='species')
plt.show()
```



[ ]:
```
#Q9. Visualize the data using 'histogram'
df.hist(figsize=(10,8))
plt.show()
```



[11]:
```
#Q10. display the first 5 objects

df.head()
```

[11]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

  species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

[12]:
```python
#Q11. display the last 5 objects

df.tail()
```

[12]:
```
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
145                6.7               3.0                5.2               2.3
146                6.3               2.5                5.0               1.9
147                6.5               3.0                5.2               2.0
148                6.2               3.4                5.4               2.3
149                5.9               3.0                5.1               1.8

       species
145  virginica
146  virginica
147  virginica
148  virginica
149  virginica
```

[13]:
```python
#Q12. display the object starting from ith to Jth position, where i and j are␣
↪parameters

def display_i_to_j(i, j):
    return df.iloc[i:j]

display_i_to_j(10, 20)
```

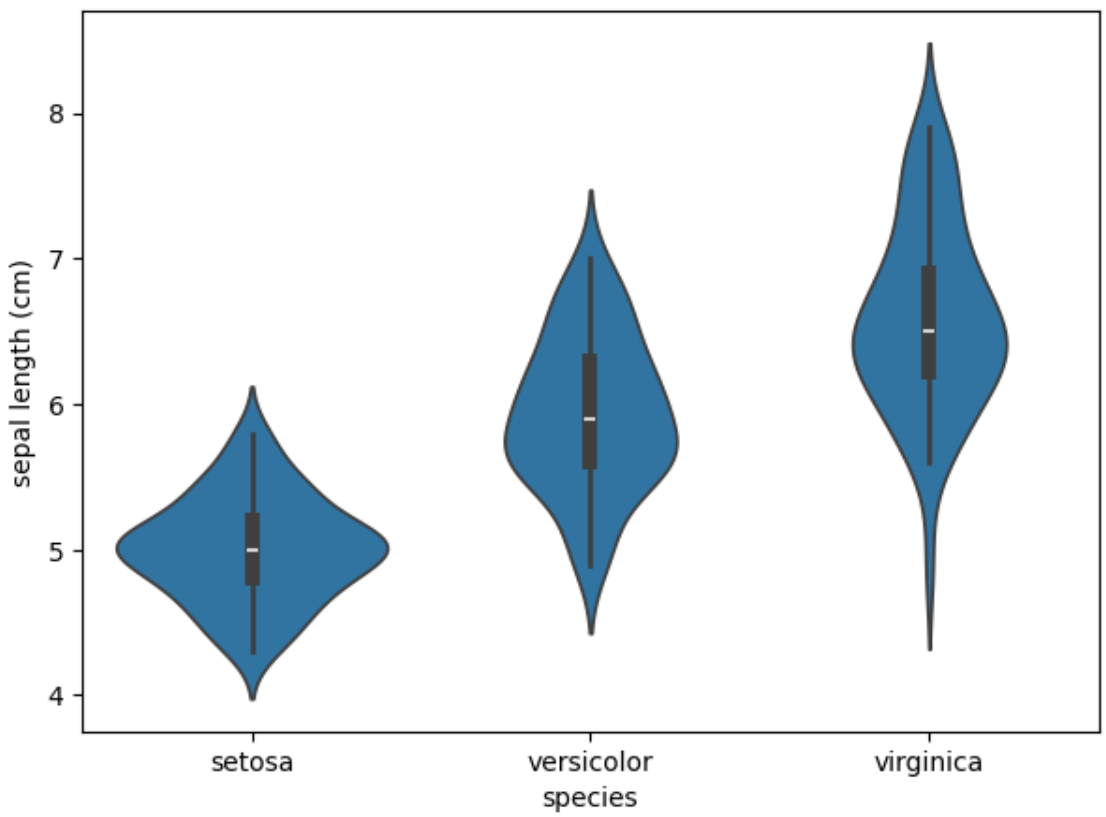
[13]:
```
    sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
10                5.4               3.7                1.5               0.2
11                4.8               3.4                1.6               0.2
12                4.8               3.0                1.4               0.1
13                4.3               3.0                1.1               0.1
14                5.8               4.0                1.2               0.2
15                5.7               4.4                1.5               0.4
16                5.4               3.9                1.3               0.4
17                5.1               3.5                1.4               0.3
18                5.7               3.8                1.7               0.3
19                5.1               3.8                1.5               0.3

   species
10  setosa
11  setosa
12  setosa
13  setosa
14  setosa
15  setosa
16  setosa
17  setosa
```

```
18  setosa
19  setosa
```

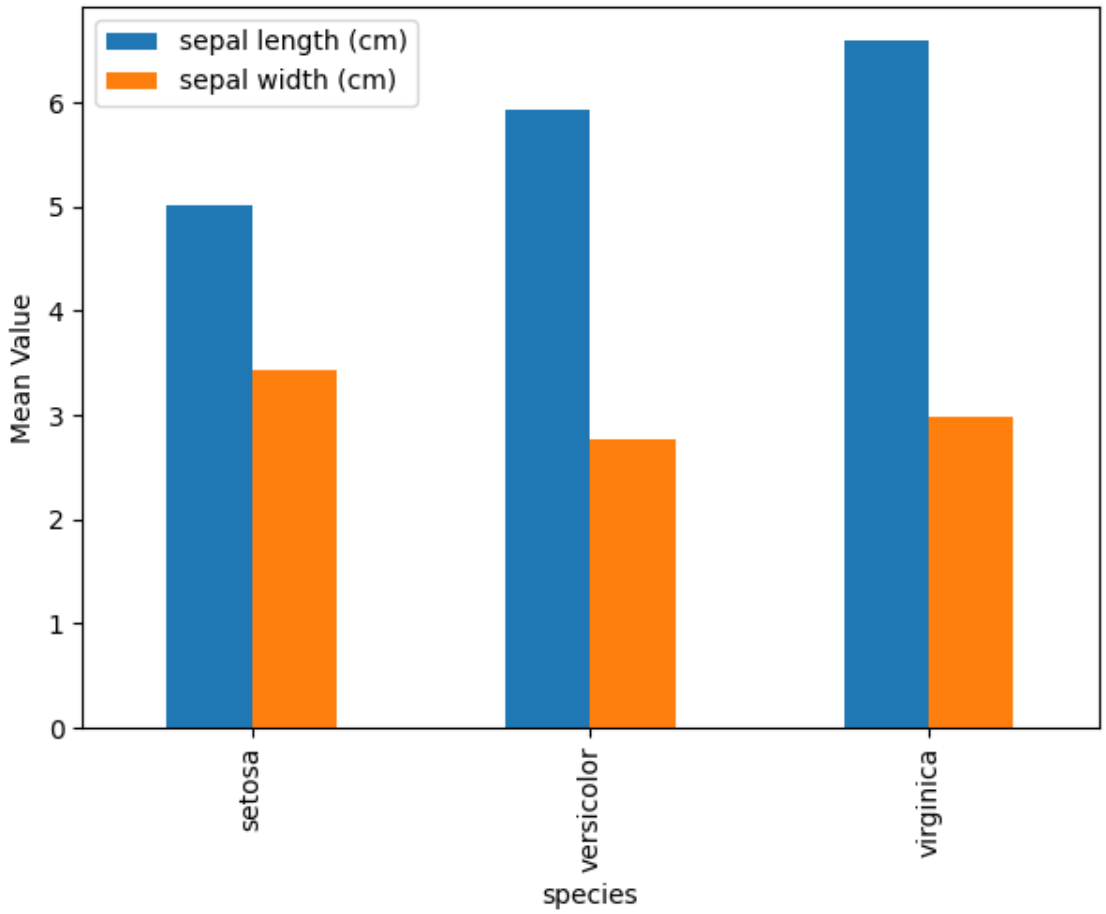
[14]:
```python
#Q13. display n objects belonging to each class, where n is a parameter

def display_n_per_class(n):
    return df.groupby('species').head(n)

display_n_per_class(3)
```

[14]:
```
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                  5.1               3.5                1.4               0.2
1                  4.9               3.0                1.4               0.2
2                  4.7               3.2                1.3               0.2
50                 7.0               3.2                4.7               1.4
51                 6.4               3.2                4.5               1.5
52                 6.9               3.1                4.9               1.5
100                6.3               3.3                6.0               2.5
101                5.8               2.7                5.1               1.9
102                7.1               3.0                5.9               2.1

        species
0        setosa
1        setosa
2        setosa
50   versicolor
51   versicolor
52   versicolor
100   virginica
101   virginica
102   virginica
```

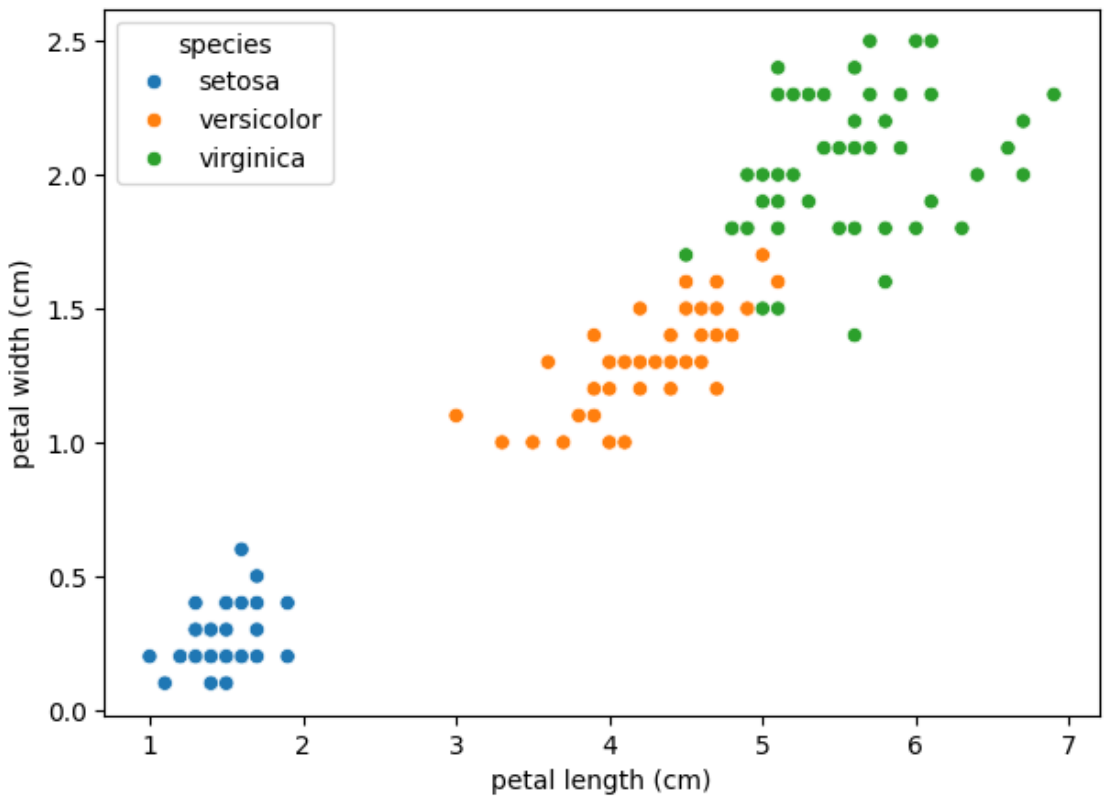
[15]:
```python
#Q14. For sepal length, display the violin plot /class

plt.figure(figsize=(7,5))
sns.violinplot(
    x='species',
    y='sepal length (cm)',
    data=df
)
plt.show()
```



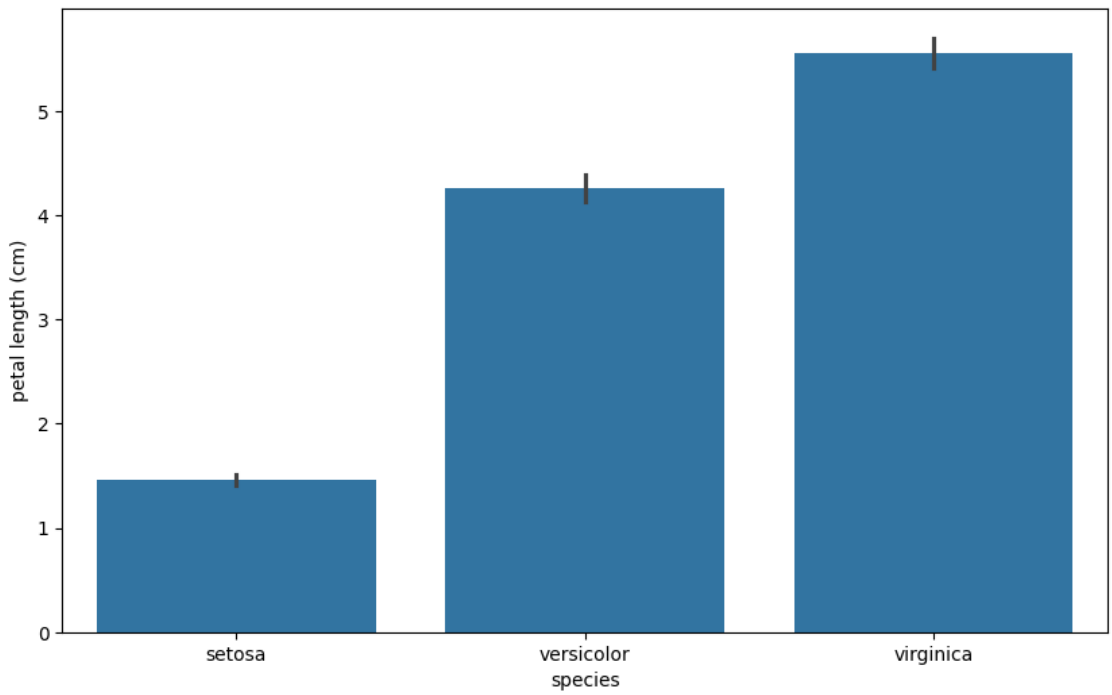
[16]:
```
#Q15. Viualize how how sepal length and sepal width vary in each class (use␣
↪barchart)

df_grouped = df.groupby('species').mean()

df_grouped[['sepal length (cm)', 'sepal width (cm)']].plot(
    kind='bar',
    figsize=(7,5)
)

plt.ylabel("Mean Value")
plt.show()
```



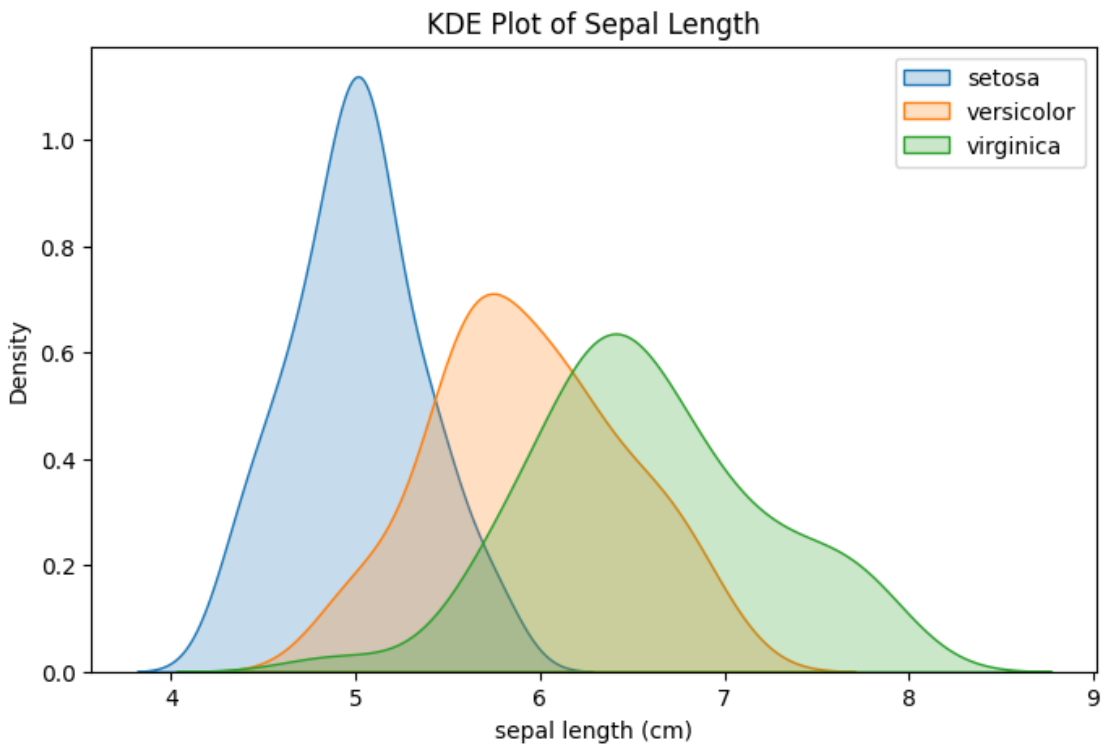
[17]:
```
#Q16. create a simple scatter plot using any two features and display the class␣
↪info using different colors

plt.figure(figsize=(7,5))
sns.scatterplot(
    data=df,
    x='petal length (cm)',
    y='petal width (cm)',
    hue='species'
)
plt.show()
```



[18]:
```
#Q17. create plots to understand differences between the species with respect␣
↪to each features(use barplot)

plt.figure(figsize=(10,6))
sns.barplot(
    data=df,
    x='species',
    y='petal length (cm)'
)
plt.show()
```



[19]:
```
#Q18. create plots to understand differences between the species with respect␣
↪to each features(use KDEplot)

plt.figure(figsize=(8,5))

for species in df['species'].unique():
    subset = df[df['species'] == species]
    sns.kdeplot(
        subset['sepal length (cm)'],
        label=species,
        fill=True
    )

plt.legend()
plt.title("KDE Plot of Sepal Length")
plt.show()
```


KDE Plot of Sepal Length
setosa
versicolor
virginica
Density
1.0
0.8
0.6
0.4
0.2
0.0
4
5
6
7
8
9
sepal length (cm)